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Supervised and Unsupervised Feature Extraction Methods for Underwater Fish Species Recognition*

Meng-Che Chuang[1], Jenq-Neng Hwang[1], Kresimir Williams[2]

[1] Department of Electrical Engineering, University of Washington
Campus Box 352500, Seattle, WA 98195, USA
{mengche, hwang}@uw.edu

[2] Alaska Fisheries Science Center, National Oceanic and Atmospheric Administration
7600 Sand Point Way NE, Seattle, WA, 98115, USA
kresimir.williams@noaa.gov

Abstract.
Automated fish species identification in open aquatic habitats based on video analytics is the primary area of research in camera-based fisheries surveys. Finding informative features for these analyses, however, is fundamentally challenging due to poor quality of underwater imagery and strong visual similarity among species. In this paper, we compare two different fish feature extraction methods, namely the supervised and unsupervised approaches, which are then applied to a hierarchical partial classification framework. Several specified anatomical parts of fish are automatically located to generate the supervised feature descriptors. For unsupervised feature extraction, a scale-invariant object part learning algorithm is proposed to discover common shape of body parts and then extract appearance, location and size information of each part. Experiments show that the unsupervised approach achieves better recognition performance on live fish images collected by trawl-based cameras.

Keywords: hierarchical classifier, partial classification, feature extraction, live fish recognition, trawl-based imagery, unsupervised learning

1 Introduction

Fish abundance estimation [6], which often calls for the use of bottom and mid-water trawls, is critically required for the conservation and management of commercially important fish populations. To improve survey quality and preserve depleted fish stocks, we developed the Cam-trawl [14] to capture image/video data of live fish passing through the trawl instead of catching them. We have developed algorithms that successfully analyze the collected data by performing fish segmentation, length estimation, counting and tracking [1], [2] and species recognition [3]. In particular,

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identification of fish species from images allows for monitoring the specie, which is essential for assessing the status of fish stocks and ecosystem.

While object recognition in various contexts has been well investigated in computer vision community, there exist fundamental challenges to identifying fish in an unconstrained natural habitat. For freely-swimming fish, there is a high uncertainty in many of the data due to poor capture quality, non-lateral fish views or curved body shapes. This seriously degrades recognition performance since some critical information may be lost. Even without uncertainty, fish share a strong visual correlation among species. Common image features for basic-level object recognition are often not sufficiently discriminative in this case.

Existing techniques of extracted fish features fall in two categories, namely supervised and unsupervised methods. Supervised methods use pre-specified features to represent a fish. Some of these features represent common image properties such as shape and texture, while others are based on domain knowledge from marine animal taxonomy. Due to the huge diversity of fish, however, a set of features designed for some species is not guaranteed to be suitable for other species. Moreover, manually selected features may lead to suboptimal recognition performance even based on fisheries knowledge. On the other hand, unsupervised methods learn informative features directly from the images. This approach has drawn emerging interest recently as a way to classify objects into subordinate categories such as bird species since it requires no prior knowledge or specification of features. An interesting question then arises: which type of approach in general leads to better recognition performance? To our best knowledge, there are yet no reported experimental results for such comparisons. A good pilot study on this is therefore highly desirable and can be used for future investigation into automatic fish species identification.

In this paper, a comparison of supervised and unsupervised fish species features for trawl-based underwater imaging is presented. We use a part-aware feature extraction focusing on specified body parts [3] as the supervised features, and propose an unsupervised scale-invariant object part model learning algorithm to discover unsupervised features. Both kinds of features are sent to a hierarchical partial classifier [3], which provides a robust solution to classifying uncertain data such as fish images.

The rest of this paper is organized as follows. Section 2 gives a brief survey on the related work. Section 3 describes the supervised method of extracting features from a fish image. Then Section 4 introduces the unsupervised feature learning method. Section 5 compares the experimental results. Finally, the conclusion is given in Section 6.

2 Related Work

There has been a large body of work proposed to recognize live fish species [8], [9], [10], [12]. Although some focus on the classification algorithms for fish species, such as Huang et al. [8], most work contributes to the design and extraction of specified features based on image analytics or computer vision techniques. Lee et al. [9] focus on local attributes including using contours or texture. Spampinato et al. [12] also combine shape and texture information and apply an affine transformation to handle view changes of the fish. Nery et al. [10] choose discriminative features for fish by traditional feature selection theories. These methods, however, do not take into account the importance of anatomical part appearance variations, which are more consistent with the knowledge from fisheries and animal taxonomy domains. In our work, part-aware features are designed to exploit discriminative information from several fish anatomical parts so that visually-similar fish species are successfully identified.

As for unsupervised feature learning methods, discovery of discriminative features for fine-grained object recognition has drawn increasing interest in recent years. For example, an annotation-free method [15] uses grid-based patches as templates and learns the template models as well as their co-occurrence relationships. Although it provides an unsupervised way to extract useful patches from images, scale variation of patches are not considered. To address this need in discovering useful patches in fish images, we propose the scale-invariant object part model for unsupervised learning of informative fish species features.

3 Supervised Feature Extraction

The major visual differences among fish species often lie in several external anatomical parts, such as the head, the caudal (tail) fin and eyes [7]. These body parts are therefore emphasized by our previous work on part-aware feature extraction to distinguish fish species. We refer to [3] for details of the method and only review the main concepts here. Image projections of fish binary masks are utilized to locate the head and tail regions. In addition, an object detector using the Viola-Jones cascade classifier [13] is trained and employed inside the head region to locate the eyes.

The size, shape and texture attributes of fish head, tail and eyes are then extracted as features for species recognition. The size attributes are estimated by the normalized area with respect to the whole body area. The shape attributes are represented by the Fourier descriptor of contour points normalized by its DC component so that the descriptor is scale-invariant. Texture properties are represented by the histogram of local binary patterns (LBP) within the detected part [11]. By leveraging fisheries knowledge, several global attributes including the aspect ratio, projection ratio and length ratio of the fish body are also added to the features, resulting in supervised fish species features with the total dimensionality 75.

4 Unsupervised Feature Learning

As discussed in Section 1, supervised feature extraction approaches have weaknesses including lack of generality and risk of suboptimal performance. In light of this, we propose an alternative feature extraction approach which systematically learns a scale-invariant object part model from images in an unsupervised fashion. An overview of the proposed unsupervised feature learning algorithm is shown in Fig. 1.

4.1 Problem formulation

Given a set of object images $\mathbf{I} = \{I^1,...,I^N\}$, the goal is to extract discriminative features from the images in terms of their subordinate categories. Let $M = \{\mathbf{P},\mathbf{X},\mathbf{S}\}$ denote a model that consists of part features $\mathbf{P} = \{\mathbf{P}_1,...,\mathbf{P}_K\}$, part center locations $\mathbf{X} = \{\mathbf{X}^1,...,\mathbf{X}^N\}$ and part sizes $\mathbf{S} = \{\mathbf{S}^1,...,\mathbf{S}^N\}$, where K is the number of parts. For each image I^m we further denote $\mathbf{X}^m = \{\mathbf{x}_1^m,...,\mathbf{x}_K^m\}$ and $\mathbf{S}^m = \{\mathbf{s}_1^m,...,\mathbf{s}_K^m\}$. The location and size of each part are represented by the normalized coordinates with respect to the image size, i.e., $x_i, s_i \in [0,1]\times[0,1]$. Based on this, discovering scale-invariant object parts can be written as a constrained minimization programming problem:

$$(\mathbf{P}^*,\mathbf{X}^*,\mathbf{S}^*) = \arg\min J(\mathbf{P},\mathbf{X},\mathbf{S}), \tag{1}$$

$$\text{s.t. } 0 \le \mathbf{s}_i \le 1,\quad i = 1,...,K, \tag{2}$$

$$0 \le \mathbf{x}_i \pm (1/2)\mathbf{s}_i \le 1,\quad i = 1,...,K, \tag{3}$$

where $J(\mathbf{P},\mathbf{X},\mathbf{S})$ is the objective function of the model. Note (2) and (3) are introduced to impose entry-wise inequalities.

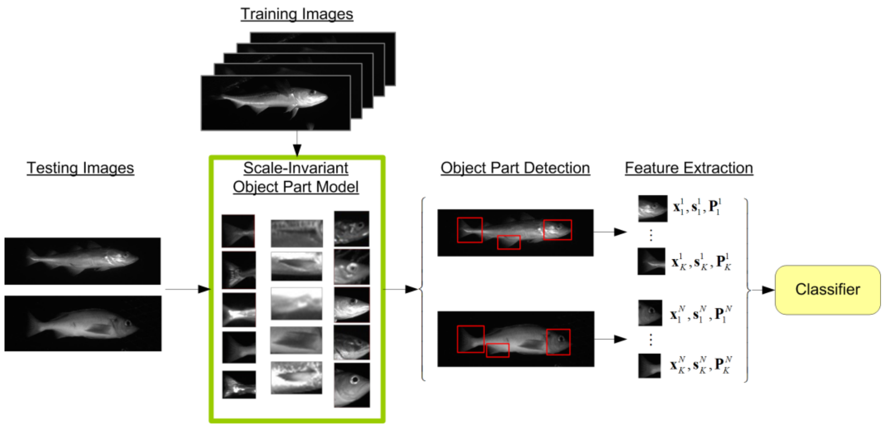

Fig. 1. Overview of the unsupervised feature extraction algorithm. In the training stage, the scale-invariant object part models are learned from training images. In the testing stage, the model locates useful object parts in each testing image. The location, size and appearance of each part are extracted as features. Finally, these features are used to train a fish classifier.

4.2 Scale-Invariant Part Learning

To model both the appearance and geometry of object parts, the objective function $J(\mathbf{P},\mathbf{X},\mathbf{S})$ takes three factors into account: 1) *fitness*, which computes the appearance similarity between the model and an image region; 2) *separation*, which guides the patches to match as disjoint as possible regions of the image instead of concentrating on a few regions of the image; 3) *discrimination*, which encourages the selected patches to have distinct appearance from each other in order to capture all aspects of the object appearance. Following these guidelines, we define the objective function as

$$J(\mathbf{P},\mathbf{X},\mathbf{S}) = J_{fitness} + J_{separation} + J_{discrimination}$$
$$= \sum_{m=1}^{N}\sum_{i=1}^{K} d(\mathbf{P}_i, I^m(\mathbf{x}_i^m,\mathbf{s}_i^m)) + \sum_{m=1}^{N}\sum_{i=1}^{K}\sum_{j\neq i} q(\mathbf{x}_i^m,\mathbf{s}_i^m,\mathbf{x}_j^m,\mathbf{s}_j^m) - \sum_{i=1}^{K}\sum_{j=1}^{K} d(\mathbf{P}_i,\mathbf{P}_j), \tag{4}$$

where d is the distance metric of appearance features and q is the normalized overlapping area of two image regions $(\mathbf{x}_i^m,\mathbf{s}_i^m)$ and $(\mathbf{x}_j^m,\mathbf{s}_j^m)$.

The scale-invariant object part model $M^* = \{\mathbf{P}^*,\mathbf{X}^*,\mathbf{S}^*\}$ is learned by solving the optimization problem (1)-(3). Here we employ an unsupervised EM-like algorithm, namely the alternating optimization. Specifically, in each iteration the algorithm first updates the locations $\mathbf{X}$ (part localization) with the rest of variables being fixed:

$$\min_{\mathbf{X}} \sum_{m=1}^{N}\left(\sum_{i=1}^{K} d(\mathbf{P}_i, I^m(\mathbf{x}_i^m,\mathbf{s}_i^m)) + \sum_{i=1}^{K}\sum_{j\neq i} q(\mathbf{x}_i^m,\mathbf{s}_i^m,\mathbf{x}_j^m,\mathbf{s}_j^m)\right), \tag{5}$$
$$\text{s.t.} \quad 0 \le \mathbf{x}_i \pm (1/2)\mathbf{s}_i \le 1, \;\; i = 1,\ldots,K. \tag{6}$$

Then the sizes $\mathbf{S}$ are updated (part size fitting) with the rest of variables being fixed:

$$\min_{\mathbf{S}} \sum_{m=1}^{N}\left(\sum_{i=1}^{K} d(\mathbf{P}_i, I^m(\mathbf{x}_i^m,\mathbf{s}_i^m)) + \sum_{i=1}^{K}\sum_{j=1}^{K} q(\mathbf{x}_i^m,\mathbf{s}_i^m,\mathbf{x}_j^m,\mathbf{s}_j^m)\right) \tag{7}$$
$$\text{s.t.} \quad 0 \le \mathbf{s}_i \le 1, \;\; i = 1,\ldots,K. \tag{8}$$

Finally, $\mathbf{P}$ is updated (part model learning) with the rest of variables being fixed:

$$\min_{\mathbf{P}_i} \sum_{m=1}^{N} d(\mathbf{P}_i, I^m(\mathbf{x}_i^m,\mathbf{s}_i^m)) - \sum_{j=1}^{K} d(\mathbf{P}_i,\mathbf{P}_j). \tag{9}$$

In part localization and size fitting steps, the mean-shift algorithm [4] is adopted to find the optimal solution efficiently. For part model learning, one can formulate (9) as a classic linear programming problem by choosing normalized correlation as the distance metric (see Section 5.2). The algorithm is repeated until the model converges.

5 Experimental Results

5.1 Hierarchical Partial Classifier

In our experiments, live fish images are classified by our previous work of hierarchical partial classification framework [3], which has shown its capability of handling data with high uncertainty and class imbalance. First, a binary species tree is learned from the data via unsupervised clustering. Each tree node represents a binary classifier that sorts data into one of its children groupings. The notion of partial classification, i.e., allowing indecision in certain regions in the data space, is introduced to each of the classifiers. Any ambiguous instance can then be assigned partial (high-level) labels. This effectively reduces misclassification while gathering much information from the uncertain data. Furthermore, the exponential benefit function is defined to evaluate each partial classifier, and hence the decision criteria for classifiers are optimized in the sense of information retrieval efficiency.

5.2 Dataset and Simulation Setup

We test the two aforementioned feature extraction methods with a self-collected underwater fish dataset. The images are captured by the Cam-trawl system [14] from a mid-water trawl with artificial LED lighting and only simple web patterns in the background. We apply our previous work on automatic fish segmentation [1] to generate reliable object bounding boxes and segmentation masks. The resultant dataset contains 1325 grayscale fish images from 7 species. Before the analysis, each image is scaled to no larger than 300×300 pixels with its aspect ratio preserved.

A 10-fold cross-validation is performed on the dataset during classifier training. All parameter settings for supervised features follow our previous work [3]. As for unsupervised features, we test the initial settings with 4, 6, 8 and 10 parts, each of which is placed as shown in Fig. 2. These initial parts are placed with an intention to efficiently discover prominent fish body parts, such as the head, caudal fin, dorsal fin and so on. The histogram of oriented gradient (HOG) [5] is used to represent the body part appearance. We choose 4×4 grid cells, 2×2 cells in one block and 1 cell of spacing as the parameters in HOG calculation. As a result, the feature dimensionality is 292 for each body part. The normalized correlation is selected as the similarity metric of HOG features, i.e., $d(\mathbf{P},\mathbf{Q}) = 1 - \mathbf{p}^T\mathbf{q}$, where $\mathbf{p} = \mathbf{P}/\|\mathbf{P}\|$ and $\mathbf{q} = \mathbf{Q}/\|\mathbf{Q}\|$.

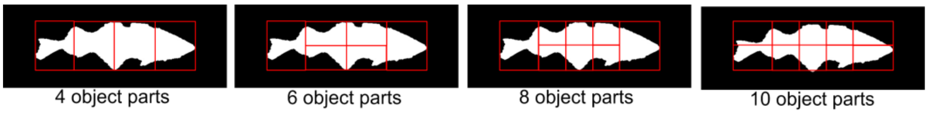

Fig. 2. Choices of initial object body part number and placement.

5.3 Recognition Performance

Fish species recognition experiments are conducted over two feature extraction methods and several feature combinations. Supervised methods include part-aware features (PAF) using all specified features, excluding length ratio, tail texture and eye texture. The unsupervised methods include the scale-invariant object part learning method (SIOPL) using 4, 6, 8 and 10 initial parts as described in Section 5.2. The accuracy and partial decision rates derived from analyzing the live fish dataset using hierarchical partial classifier are shown in Table 1. All unsupervised approaches outperform the supervised ones with a margin of at least 5%. This implies that the unsupervised object part discovery exploits some discriminative information that is not considered in the supervised features. Note that the partial decision rates are below 6% for all cases, showing that the hierarchical partial classifier can achieve favorable recognition without discarding a large portion of information from uncertain data. The precision, recall and F_1-score of each fish species are shown in Table 2 as a more complete report of the performance. We refer to [3] for definitions of these metrics.

Fig. 3 gives a visualization of some fish body parts learned and detected by the proposed scale-invariant object part learning algorithm. In addition to the head and the caudal fin, which are considered in both types of features, the unsupervised algorithm locates the pectoral fin and classify the fish based on its characteristics. If the pectoral fin features are removed deliberately, the recognition performance is decreased, as shown in the last row of Table 1. This serves as a justification for the high impact of salient features systematically discovered by the unsupervised approach.

Table 1. Accuracy and partial decision rates vs. feature extraction methods

Feature Type	Accu. (%)	PD (%)	Feature Type	Accu. (%)	Indec. (%)
PAF-all	93.65	4.91	SIOPL-4	98.94	1.06
PAF-noLenRatio	84.48	3.32	SIOPL-6	99.32	5.84
PAF-noTailTex	74.66	2.98	SIOPL-8	99.92	3.92
PAF-noEyeTex	81.76	4.27	SIOPL-10	98.43	4.31
			SIOPL-8-noPec	94.15	4.09

Table 2. Precision, recall and F_1-score (%) of each fish species by using SIOPL-8 features

Species	King Salmon	Chum Salmon	Juvenile Pollock	Adult Pollock	Capelin	Eulachon	Rockfish
Precision	100	100	99.3	100	100	100	100
Recall	100	100	100	100	100	99.5	100
F_1-score	100	100	99.6	100	100	99.8	100

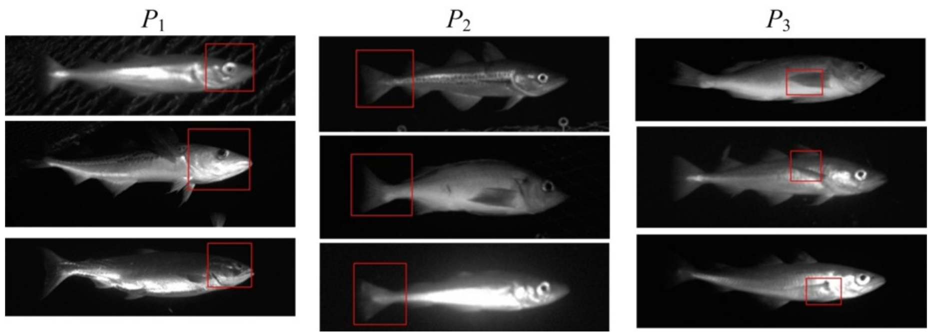

Fig. 3. Examples of parts detected by the proposed unsupervised feature learning algorithm. Each column shows the parts found by one part model. Meaningful fish body parts are successfully learned regardless changes in size, such as head, caudal fin and pectoral fin.

6 Conclusion

In this paper, an experimental comparison is presented between two different fish feature extraction methods. Even though the supervised method defines specified fish body parts based on domain knowledge, such feature descriptors may fail to exploit subtle but useful characteristics. In contrast, the unsupervised scale-invariant object part model discovers features in a general way and thus extracts subtle variations among fish species. Experiments verify that the unsupervised method achieves a better performance with live fish images where image quality is an issue and the species distribution is highly imbalanced. Future work includes testing the unsupervised approach with different datasets and incorporating temporal information from video.

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